

Entropic Methods and Wasserstein Distances

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1 Introduction

These notes correspond to Sections 3.1.2 and 3.3.1 of [1]. There are two topics.

- (i) **The Herbst argument.** We previously introduced the entropy functional $H(Z)$. The Herbst argument shows that a bound on the entropy $H(e^{\lambda X})$ gives a sub-Gaussian bound on the cumulant generating function. By the Chernoff method, this then gives a tail bound.
- (ii) **Wasserstein distances.** We define the Wasserstein distance through Lipschitz test functions and show that the Hamming metric gives the total variation distance. We then relate the Wasserstein distance to optimal transport through couplings.

2 Background: the Entropy Functional

Definition 2.1 (ϕ -entropy). For a convex function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ and $X \sim P$, define

$$H_\phi(X) := \mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X]).$$

By Jensen's inequality, $H_\phi(X) \geq 0$. It measures how much X varies. For example, $\phi(u) = u^2$ gives $H_\phi(X) = \text{Var}(X)$.

Throughout, we take the convex function $\phi : [0, \infty) \rightarrow \mathbb{R}$ given by $\phi(u) = u \log u$ for $u > 0$ and $\phi(0) = 0$. For $Z \geq 0$ we write

$$H(Z) = \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z]. \quad (1)$$

Lemma 2.2 (Entropy of $e^{\lambda X}$ via the MGF). *Let $\varphi_X(\lambda) = \mathbb{E}[e^{\lambda X}]$ be the moment generating function (MGF) of X , and assume it is finite near the relevant λ , so that we may differentiate under the expectation. Then*

$$H(e^{\lambda X}) = \lambda \varphi'_X(\lambda) - \varphi_X(\lambda) \log \varphi_X(\lambda). \quad (2)$$

Proof. Differentiating under the expectation gives $\varphi'_X(\lambda) = \mathbb{E}[Xe^{\lambda X}]$. With $Z = e^{\lambda X}$ we have $Z \log Z = \lambda X e^{\lambda X}$, so $\mathbb{E}[Z \log Z] = \lambda \varphi'_X(\lambda)$ and $\mathbb{E}[Z] = \varphi_X(\lambda)$. Substituting into (1) gives (2). $\square$

Example 2.3 (Gaussian case: equality). If $X \sim N(0, \sigma^2)$, then $\varphi_X(\lambda) = e^{\lambda^2 \sigma^2 / 2}$ and $\varphi'_X(\lambda) = \lambda \sigma^2 \varphi_X(\lambda)$. Hence, by (2),

$$H(e^{\lambda X}) = \lambda^2 \sigma^2 \varphi_X(\lambda) - \frac{1}{2} \lambda^2 \sigma^2 \varphi_X(\lambda) = \frac{1}{2} \lambda^2 \sigma^2 \varphi_X(\lambda).$$

So the hypothesis of the Herbst argument below holds *with equality* for Gaussians. This is the motivating example.

Remark 2.4. By [1, Exercise 3.7], if X is zero-mean and $X \in [a, b]$ almost surely, then $H(e^{\lambda X}) \leq \frac{1}{2} \sigma^2 \lambda^2 \varphi_X(\lambda)$ with $\sigma = (b - a)/2$. So bounded random variables also satisfy the entropy condition.

3 The Herbst Argument

Proposition 3.1 (Herbst argument). *Let $\sigma > 0$ be a constant, $\varphi_X(\lambda) = \mathbb{E}[e^{\lambda X}]$, and $I = \mathbb{R}$ or $I = [0, +\infty)$. Suppose the entropy $H(e^{\lambda X})$ satisfies*

$$H(e^{\lambda X}) \leq \frac{1}{2}\sigma^2\lambda^2\varphi_X(\lambda) \quad \text{for all } \lambda \in I. \quad (3)$$

Then X satisfies

$$\log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \frac{1}{2}\lambda^2\sigma^2 \quad \text{for all } \lambda \in I, \quad (4)$$

or equivalently, $\mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq e^{\frac{\lambda^2\sigma^2}{2}}$ for all $\lambda \in I$.

Proof of Proposition 3.1. Case 1: $I = [0, \infty)$.

① **A differential inequality for the MGF.** Combining (2) with the hypothesis (3) gives

$$\lambda\varphi'_X(\lambda) - \varphi_X(\lambda) \log \varphi_X(\lambda) \leq \frac{1}{2}\sigma^2\lambda^2\varphi_X(\lambda) \quad \text{for all } \lambda \in I. \quad (5)$$

Since $\varphi_X(\lambda) > 0$, we can divide by $\varphi_X(\lambda)$:

$$\frac{\lambda\varphi'_X(\lambda)}{\varphi_X(\lambda)} - \log \varphi_X(\lambda) \leq \frac{1}{2}\sigma^2\lambda^2, \quad \lambda \geq 0. \quad (6)$$

② **The auxiliary function G .** The key step is to rewrite the left side of (6) as a derivative. Define

$$G(\lambda) := \frac{\log \varphi_X(\lambda)}{\lambda}, \quad \lambda > 0.$$

By the product rule,

$$G'(\lambda) = -\frac{1}{\lambda^2} \log \varphi_X(\lambda) + \frac{1}{\lambda} \cdot \frac{\varphi'_X(\lambda)}{\varphi_X(\lambda)} = \frac{1}{\lambda^2} \left[\frac{\lambda\varphi'_X(\lambda)}{\varphi_X(\lambda)} - \log \varphi_X(\lambda) \right].$$

So, up to the factor $1/\lambda^2$, $G'(\lambda)$ is exactly the left side of (6). Dividing (6) by $\lambda^2 > 0$ gives

$$G'(\lambda) \leq \frac{1}{2}\sigma^2 \quad \text{for all } \lambda > 0. \quad (7)$$

③ **Integrate.** Fix $\lambda > 0$. For any $\lambda_0 \in (0, \lambda]$, by the fundamental theorem of calculus and (7),

$$G(\lambda) - G(\lambda_0) = \int_{\lambda_0}^{\lambda} G'(s) ds \leq \int_{\lambda_0}^{\lambda} \frac{1}{2}\sigma^2 ds = \frac{1}{2}\sigma^2(\lambda - \lambda_0).$$

(We need $\lambda_0 \leq \lambda$ here: if $\lambda < \lambda_0$, the integral runs backwards and the inequality flips. This does no harm, because we next send $\lambda_0 \rightarrow 0^+$.)

④ **Let $\lambda_0 \rightarrow 0^+$.** Taking the limit (which exists, as computed below),

$$G(\lambda) - \lim_{\lambda_0 \rightarrow 0^+} G(\lambda_0) \leq \frac{1}{2}\sigma^2\lambda.$$

To compute the limit, use $\varphi_X(0) = \mathbb{E}[e^0] = 1$, so $\log \varphi_X(0) = 0$. The limit is then a right-hand difference quotient:

$$\begin{aligned} \lim_{\lambda_0 \rightarrow 0^+} G(\lambda_0) &= \lim_{\lambda_0 \rightarrow 0^+} \frac{\log \varphi_X(\lambda_0)}{\lambda_0} = \lim_{\lambda_0 \rightarrow 0^+} \frac{\log \varphi_X(\lambda_0) - \log \varphi_X(0)}{\lambda_0 - 0} = \frac{d}{d\lambda} \log \varphi_X(\lambda) \Big|_{\lambda=0} \\ &= \frac{\varphi'_X(0)}{\varphi_X(0)} = \frac{\mathbb{E}[X]}{1} = \mathbb{E}[X]. \end{aligned}$$

So G extends continuously to 0 with $G(0) := \mathbb{E}[X]$.

⑤ **Unwind.** We have shown that for all $\lambda > 0$,

$$G(\lambda) - \mathbb{E}[X] \leq \frac{1}{2}\sigma^2\lambda \iff \frac{\log \varphi_X(\lambda)}{\lambda} - \mathbb{E}[X] \leq \frac{1}{2}\sigma^2 \iff \log \varphi_X(\lambda) - \lambda \mathbb{E}[X] \leq \frac{1}{2}\sigma^2\lambda^2.$$

Since $\log \varphi_X(\lambda) - \lambda \mathbb{E}[X] = \log \mathbb{E}[e^{\lambda X}] - \log e^{\lambda \mathbb{E}[X]} = \log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}]$, this is

$$\log \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \frac{1}{2}\sigma^2\lambda^2 \quad \text{for all } \lambda > 0.$$

The bound also holds at $\lambda = 0$, where both sides equal 0. This proves the claim for $I = [0, \infty)$.

Case 2: $I = \mathbb{R}$. The text leaves this case as an exercise. For $\lambda < 0$, (5) still holds. Dividing by $\lambda^2 \varphi_X(\lambda) > 0$ again gives $G'(\lambda) \leq \frac{1}{2}\sigma^2$ for all $\lambda < 0$. Fix $\lambda < \lambda_0 < 0$ and integrate:

$$G(\lambda_0) - G(\lambda) \leq \frac{1}{2}\sigma^2(\lambda_0 - \lambda).$$

Let $\lambda_0 \rightarrow 0^-$. The same difference-quotient argument gives $G(\lambda_0) \rightarrow \mathbb{E}[X]$, so $\mathbb{E}[X] - G(\lambda) \leq -\frac{1}{2}\sigma^2\lambda$, that is, $G(\lambda) \geq \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda$. Multiplying by $\lambda < 0$ *reverses* the inequality:

$$\log \varphi_X(\lambda) = \lambda G(\lambda) \leq \lambda \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda^2,$$

which is (4) for $\lambda < 0$. □

Remark 3.2 (Why G ?). The function $\lambda G(\lambda) = \log \varphi_X(\lambda)$ is the cumulant generating function. The target (4) reads $\log \varphi_X(\lambda) \leq \lambda \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda^2$, which is $G(\lambda) \leq \mathbb{E}[X] + \frac{1}{2}\sigma^2\lambda$. So it is enough to show that G starts at $\mathbb{E}[X]$ and has slope at most $\sigma^2/2$. This reduces the problem to a one-line integration.

Remark 3.3 (Interpretation and consequences). (a) If $I = \mathbb{R}$, then (4) says exactly that $X - \mathbb{E}[X]$ is *sub-Gaussian with parameter σ* .

(b) If $I = [0, \infty)$, the Chernoff method still gives the one-sided bound: for every $t \geq 0$,

$$\mathbb{P}[X \geq \mathbb{E}[X] + t] \leq \inf_{\lambda \geq 0} e^{-\lambda t} \mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \inf_{\lambda \geq 0} e^{-\lambda t + \frac{\lambda^2 \sigma^2}{2}} = e^{-\frac{t^2}{2\sigma^2}},$$

where the infimum is attained at $\lambda^* = t/\sigma^2$.

(c) If $I = \mathbb{R}$, applying the same bound to $-X$ gives $\mathbb{P}[|X - \mathbb{E}[X]| \geq t] \leq 2e^{-\frac{t^2}{2\sigma^2}}$.

In short, *entropy control* $\Rightarrow$ *MGF control* $\Rightarrow$ *tail control*. The entropy method is useful because entropy *tensorizes* over independent coordinates, which the MGF does not do as easily.

4 Wasserstein Distances

This section follows Section 3.3.1 of [1]. The material leads into optimal transport and transportation cost inequalities.

Definition 4.1 (Lipschitz functions). Let $(\mathcal{X}, \rho)$ be a metric space. A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is *L-Lipschitz with respect to ρ* if

$$|f(x) - f(x')| \leq L \cdot \rho(x, x') \quad \text{for all } x, x' \in \mathcal{X}.$$

We write $\|f\|_{\text{Lip}}$ for the smallest L for which this holds:

$$\|f\|_{\text{Lip}} = \sup_{x \neq x'} \frac{|f(x) - f(x')|}{\rho(x, x')}.$$

Definition 4.2 (Wasserstein distance induced by ρ). Given two probability distributions Q, P on $\mathcal{X}$, define

$$W_\rho(Q, P) := \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f dQ - \int f dP \right). \quad (8)$$

This is the *Wasserstein distance induced by ρ* (also called the W_1 or Kantorovich–Rubinstein distance).

Remark 4.3 (Invariance to constant offsets). For any constant c , $\int (f+c) dQ - \int (f+c) dP = \int f dQ - \int f dP$, because both measures have total mass 1. So in (8) we can shift each f by any constant without changing the value.

5 Example: Hamming Metric and Total Variation

Consider the *Hamming (discrete) metric*

$$\rho(x, x') = \mathbf{1}_{\{x \neq x'\}}.$$

Throughout, all sets and functions are measurable.

Definition 5.1 (Three definitions of total variation). For probability distributions Q, P on $\mathcal{X}$:

- (a) (Sets) $\|Q - P\|_{\text{TV}} = \sup_{A \subseteq \mathcal{X}} |Q(A) - P(A)|$.
- (b) (Test functions) $\|Q - P\|_{\text{TV}} = \sup_{f: \mathcal{X} \rightarrow [0,1]} \left(\int f dQ - \int f dP \right)$.
- (c) (Densities) If P and Q have densities p and q with respect to a base (dominating) measure ν ,

$$\|Q - P\|_{\text{TV}} = \frac{1}{2} \int |p(x) - q(x)| \nu(dx).$$

Remark 5.2. A dominating measure always exists: take $\nu = \frac{1}{2}(P + Q)$, so that both $P, Q \ll \nu$. The value in Definition 5.1(c) does not depend on which dominating ν is chosen.

Proposition 5.3. For the Hamming metric, W_ρ equals the total variation distance:

$$W_{\text{Ham}}(Q, P) = \|Q - P\|_{\text{TV}} := \sup_{A \subseteq \mathcal{X}} |Q(A) - P(A)|,$$

where the supremum ranges over all measurable A .

Proof of Proposition 5.3. Key observation. If $\|f\|_{\text{Lip}} \leq 1$ with respect to the Hamming metric, then $|f(x) - f(x')| \leq 1 \cdot \rho(x, x') \leq 1$ for all x, x' . Conversely, any $f: \mathcal{X} \rightarrow [0, 1]$ satisfies $|f(x) - f(x')| \leq \mathbf{1}_{\{x \neq x'\}}$, so it is 1-Lipschitz.

Reduction to $[0, 1]$ -valued functions. By Remark 4.3, W_ρ does not change if f is shifted by a constant. If $\|f\|_{\text{Lip}} \leq 1$, the key observation gives $\sup f - \inf f \leq 1$, so $\tilde{f} := f - \inf f$ takes values in $[0, 1]$ and gives the same objective. Hence we may restrict the supremum to functions with $f(x) \in [0, 1]$ for all $x \in \mathcal{X}$:

$$\begin{aligned} W_{\text{Ham}}(Q, P) &= \sup_{\substack{\|f\|_{\text{Lip}} \leq 1 \\ f: \mathcal{X} \rightarrow [0,1]}} \left(\int f dQ - \int f dP \right) \\ &= \sup_{f: \mathcal{X} \rightarrow [0,1]} \left(\int f dQ - \int f dP \right) && \text{(key observation)} \\ &= \|Q - P\|_{\text{TV}} && ([1, \text{Exercise 3.13}]). \end{aligned}$$

The second line is exactly Definition 5.1(b), and the last equality is the equivalence of Definitions 5.1(a) and 5.1(b) (homework #2). $\square$

Remark 5.4. Putting everything together, for the Hamming metric

$$W_{\text{Ham}}(Q, P) = \sup_{f: \mathcal{X} \rightarrow [0,1]} \int f(dQ - dP) = \|Q - P\|_{\text{TV}} = \frac{1}{2} \int |p - q| d\nu = \inf_M M[X \neq X'].$$

Here the equality with $\frac{1}{2} \int |p - q| d\nu$ is Definition 5.1(c), and the coupling form $\inf_M M[X \neq X']$ is also part of [1, Exercise 3.13]. In words, W_{Ham} is one half of the $L^1(\nu)$ distance between the densities. The last equality is a concrete instance of the Kantorovich–Rubinstein duality in the next section.

6 Optimal Transport and Couplings

Definition 6.1 (Coupling). A distribution M on $\mathcal{X} \otimes \mathcal{X}$ is a *coupling* of the pair (Q, P) if its marginal distributions in the first and second coordinates are Q and P respectively:

$$M(A \times \mathcal{X}) = Q(A), \quad M(\mathcal{X} \times B) = P(B) \quad \text{for all } A, B.$$

Equivalently, $(X, X') \sim M$ with $X \sim Q$ and $X' \sim P$. Couplings always exist; for example, the product $M = Q \otimes P$.

Transport cost. The quantity

$$\inf_{\substack{M \text{ a coupling} \\ \text{of } (Q, P)}} \mathbb{E}_M [\rho(X, X')], \quad (X, X') \sim M,$$

is also a distance between Q and P . It is the minimal expected cost of moving mass from Q to P when moving a unit of mass from x to x' costs $\rho(x, x')$.

6.1 Weak Duality

Proposition 6.2 (Weak duality: $\inf \geq \sup$). Assume Q and P have finite first moments. Then

$$\inf_{\substack{M \text{ coupling} \\ \text{of } (Q, P)}} \int \rho(x, x') dM(x, x') \geq \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\int f dQ - \int f dP \right) = W_\rho(Q, P). \quad (9)$$

Proof. Every 1-Lipschitz f is integrable, since $|f(x)| \leq |f(x_0)| + \rho(x, x_0)$. Fix any f with $\|f\|_{\text{Lip}} \leq 1$ and any coupling M of (Q, P) . Then

$$\begin{aligned} \int \rho(x, x') dM(x, x') &\geq \int (f(x) - f(x')) dM(x, x') && (\text{since } \|f\|_{\text{Lip}} \leq 1) \\ &= \int f(x) dM(x, x') - \int f(x') dM(x, x') && (\text{linearity}) \\ &= \int f(x) dQ(x) - \int f(x') dP(x') && (\text{marginals of } M \text{ are } Q, P) \\ &= \int f dQ - \int f dP. \end{aligned}$$

The left side does not depend on f and the right side does not depend on M . Taking the infimum over couplings on the left and the supremum over 1-Lipschitz f on the right gives (9). Kantorovich–Rubinstein duality says equality always holds $\square$

6.2 Transportation Interpretation

Suppose Q and P have densities q and p on $\mathcal{X}$, and a coupling M has density $m(x, x')$. Think of q as an initial pile of mass and p as the target configuration. Then:

- $\rho(x, x')$ is the cost per unit of moving mass from x to x' ;
- $m(x, x') dx dx'$ is the amount of mass sent from near x to near x' (a *transport plan*);
- the marginal constraints say the plan removes exactly q and delivers exactly p ;
- the total cost is $\int \int \rho(x, x') m(x, x') dx dx'$.

Minimizing over admissible plans gives the Wasserstein distance. This is why it is called the “earth mover’s” or transportation distance.

6.3 Why This Matters

The coupling form of W_ρ is the main tool for *transportation cost inequalities* ([1, Section 3.3.2]), which are bounds of the form

$$W_\rho(Q, P) \leq \sqrt{2\gamma D(Q \| P)} \quad \text{for all } Q.$$

By the dual (Lipschitz) form, such a bound gives sub-Gaussian concentration for every 1-Lipschitz f under P . Pinsker’s inequality is the Hamming case: $\|Q - P\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D(Q \| P)}$, that is, $\gamma = \frac{1}{4}$. The coupling form is also what allows these inequalities to tensorize and to handle dependent data such as Markov chains.

Summary

1. **Herbst:** $H(e^{\lambda X}) \leq \frac{1}{2} \sigma^2 \lambda^2 \varphi_X(\lambda)$ on I implies $\log \mathbb{E} [e^{\lambda(X - \mathbb{E}[X])}] \leq \frac{1}{2} \sigma^2 \lambda^2$ on I . The proof sets $G(\lambda) = \frac{1}{\lambda} \log \varphi_X(\lambda)$, shows $G' \leq \sigma^2/2$, integrates, and uses $G(0^+) = \mathbb{E}[X]$.
2. **Wasserstein:** $W_\rho(Q, P) = \sup_{\|f\|_{\text{Lip}} \leq 1} (\int f dQ - \int f dP)$, which is invariant to constant offsets of f .
3. **Hamming gives TV:**

$$W_{\text{Ham}}(Q, P) = \sup_{f: \mathcal{X} \rightarrow [0,1]} \int f (dQ - dP) = \sup_A |Q(A) - P(A)| = \frac{1}{2} \int |p - q| d\nu = \inf_M \mathbb{P}_M [X \neq X'].$$

4. **Optimal transport:** For every coupling M and 1-Lipschitz f , $\mathbb{E}_M [\rho(X, X')] \geq \int f dQ - \int f dP$, so $\inf \geq \sup$. Kantorovich–Rubinstein duality says equality always holds

References

- [1] M. J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019.
- [2] C. Villani, *Optimal Transport: Old and New*, Springer, 2009.